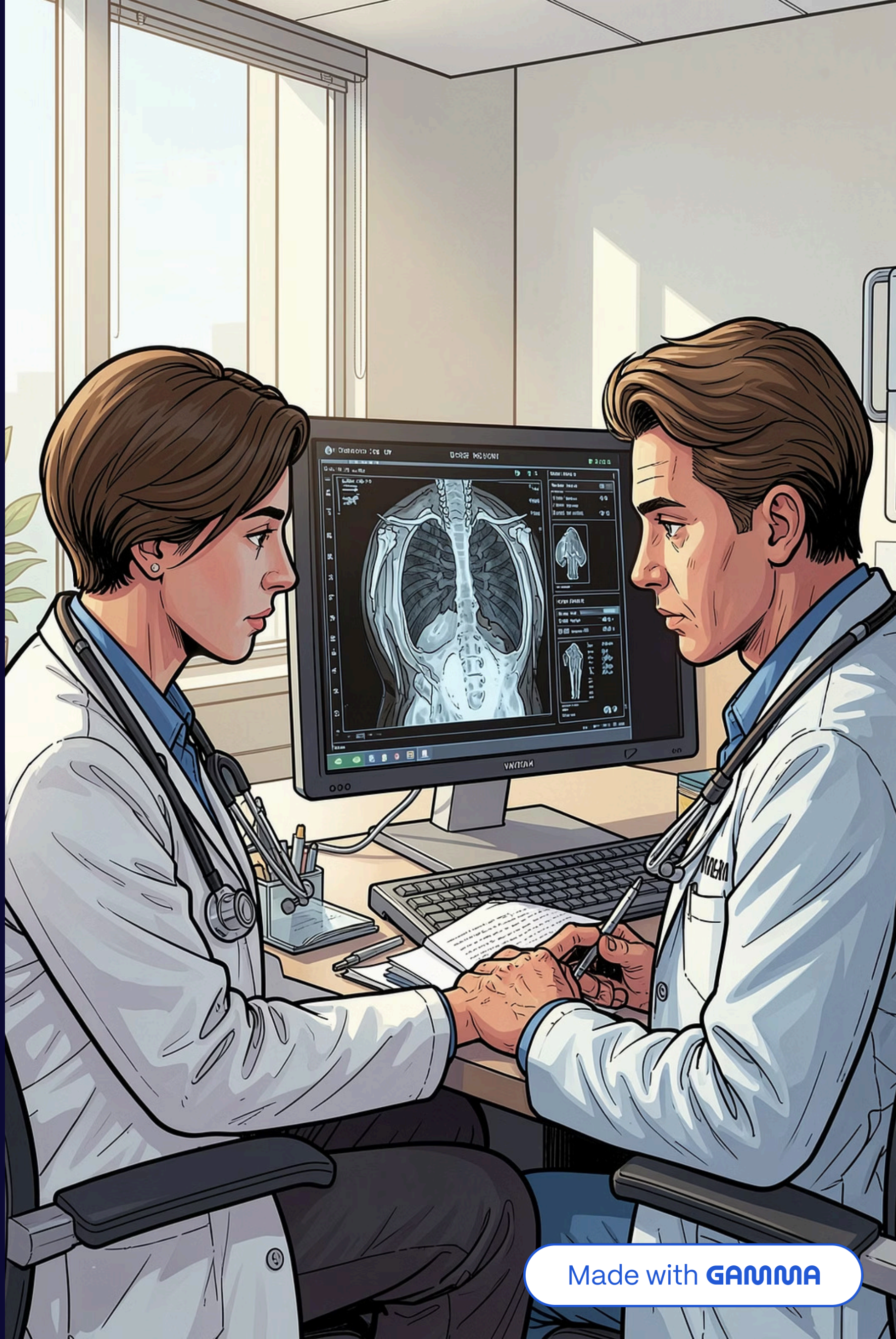
Radiomics For Early Cancer Diagnosis

Clear, practical pipeline that combines 3D CT image analysis with patient tabular data — designed for ML engineers and medical imaging researchers. This deck explains preprocessing, the 3D CNN image encoder, tabular encoder, fusion strategy, and training considerations.



Problem Statement

Lung cancer is the world's leading cause of cancer death, and survival depends critically on stage at diagnosis — five-year survival exceeds 60% when the disease is caught early but falls below 10% once it has spread. Yet only about 28% of cases are diagnosed at an early stage, while 43% are not caught until a late stage when the survival rate is only 10%. CT screening has not solved this; it has shifted the problem to one of deciding which of many indeterminate pulmonary nodules are malignant. Current computational aids are split between classical radiomics (interpretable, but dependent on manual segmentation and hand-engineered features) and deep learning (powerful, but data-hungry and opaque).

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2	LIDC-IDRI-0001					NO	3.38E+08	Homo sapi	1.3.6.1.4.1	2000-01-01	00:00:00.0	
3	LIDC-IDRI-0001					NO	3.38E+08	Homo sapi	1.3.6.1.4.1	2000-01-01	00:00:00.0	
4	LIDC-IDRI-0002					NO	3.38E+08	Homo sapi	1.3.6.1.4.1	2000-01-01	00:00:00.0	
5	LIDC-IDRI-0002					NO	3.38E+08	Homo sapi	1.3.6.1.4.1	2000-01-01	00:00:00.0	
6	LIDC-IDRI-0003					NO	3.38E+08	Homo sapi	1.3.6.1.4.1	2000-01-01	00:00:00.0	
7	LIDC-IDRI-0003					NO	3.38E+08	Homo sapi	1.3.6.1.4.1	2000-01-01	00:00:00.0	
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13	LIDC-IDRI-0006					NO	3.38E+08	Homo sapi	1.3.6.1.4.1	2000-01-01	00:00:00.0	
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28	LIDC-IDRI-0014					NO	3.38E+08	Homo sapi	1.3.6.1.4.1	2000-01-01	00:00:00.0	

Dataset

- LIDC-IDRI dataset: 152 lung CT scans in NIfTI format (ct.nii.gz)
- Each case paired with expert radiologist annotations: binary nodule masks (mask__nodule_*__reader_*__mal_*__class_*.nii.gz)
- Masks generated from LIDC XML annotations

Overview — Why rigorous preprocessing matters

Raw CTs vary by slice count, resolution, orientation, intensity scaling, and file structure. Standardization prevents downstream model confusion, enforces consistent spatial scale, and ensures that learned features reflect anatomy rather than scanner artifacts.

Clinical goal

Predict malignancy by combining radiology patterns and patient data.

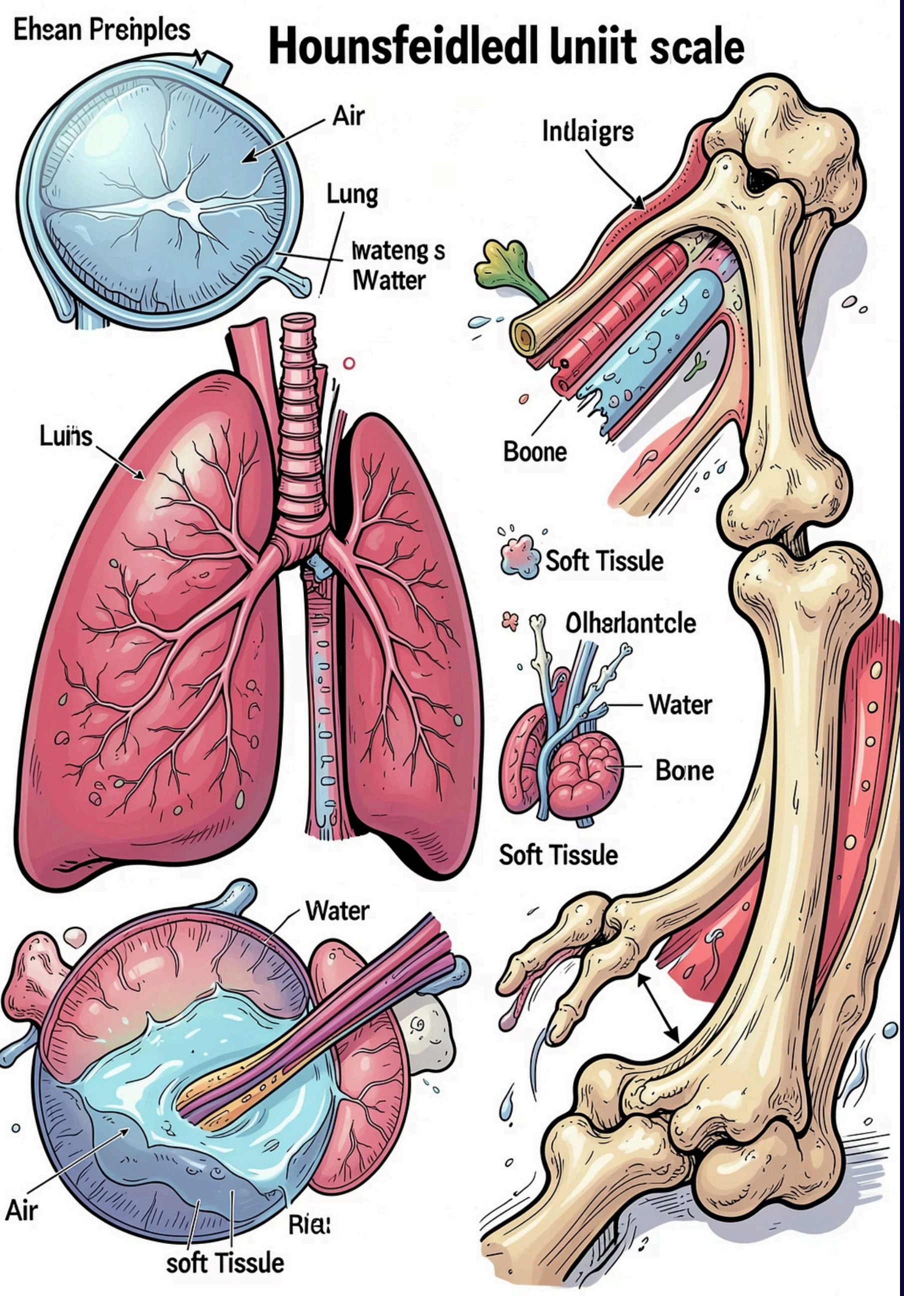
Engineering goal

Turn heterogeneous DICOMs into identical tensors for reliable training.

Part 1 — Step 1: Locate & match patient CT series

- Traverse DICOM directories, detect series using SeriesInstanceUID.
- Match series to patient IDs from Excel/registry using metadata (PatientID, StudyDate).
- Select best series (largest slice count, complete coverage).

Analogy: like indexing thousands of medical folders so each chart points to the correct scan.



Part 1 — Steps 2–3: HU conversion, clipping, normalization

Pipeline:

1. HU conversion: HU = pixel * RescaleSlope + RescaleIntercept — standardizes tissue density across scanners.
2. Clip HU range to [-1000, 400] to focus on lung and soft tissue; remove extreme bone/artifacts.
3. Normalize intensities to [0,1]: (volume - min)/(max - min) for stable network training.

Clinical reference: Air ≈ -1000 HU, Lung ≈ -700 HU, Water ≈ 0 HU, Bone ≈ +1000 HU.

Two Different Approaches

AI VS. Medical Imaging processing

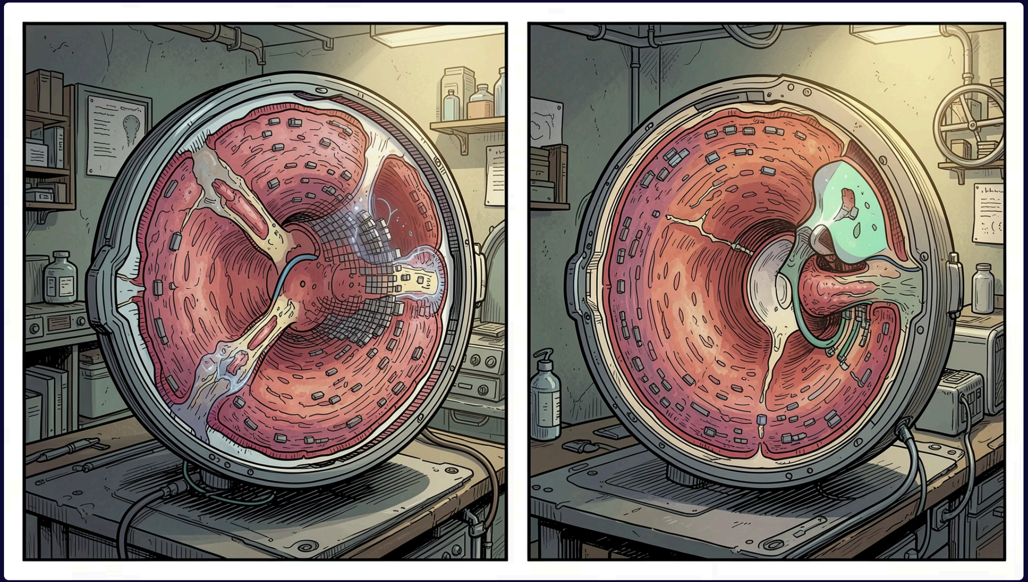
AI

Load DICOM Slices into a 3D Volume

Slices are sorted anatomically using:

- ImagePositionPatient
- InstanceNumber

Part 1 — Steps 4-5: Resample, crop/pad, final tensor



Resample: resample_volume() to target spacing (Z, Y, X) = (2.0mm, 1.0mm, 1.0mm). Use interpolation: linear for intensities; nearest for masks.

Crop/Pad: center-crop or zero-pad to (96,160,160). If Z < 96, pad symmetrically; if Z > 96, take anatomically-centered slices.

Final tensor: shape [1, 96, 160, 160] — single grayscale channel suitable for 3D CNN input.

Center Crop or Pad

Why?

Scans vary in size.

So:

Large scans → crop center

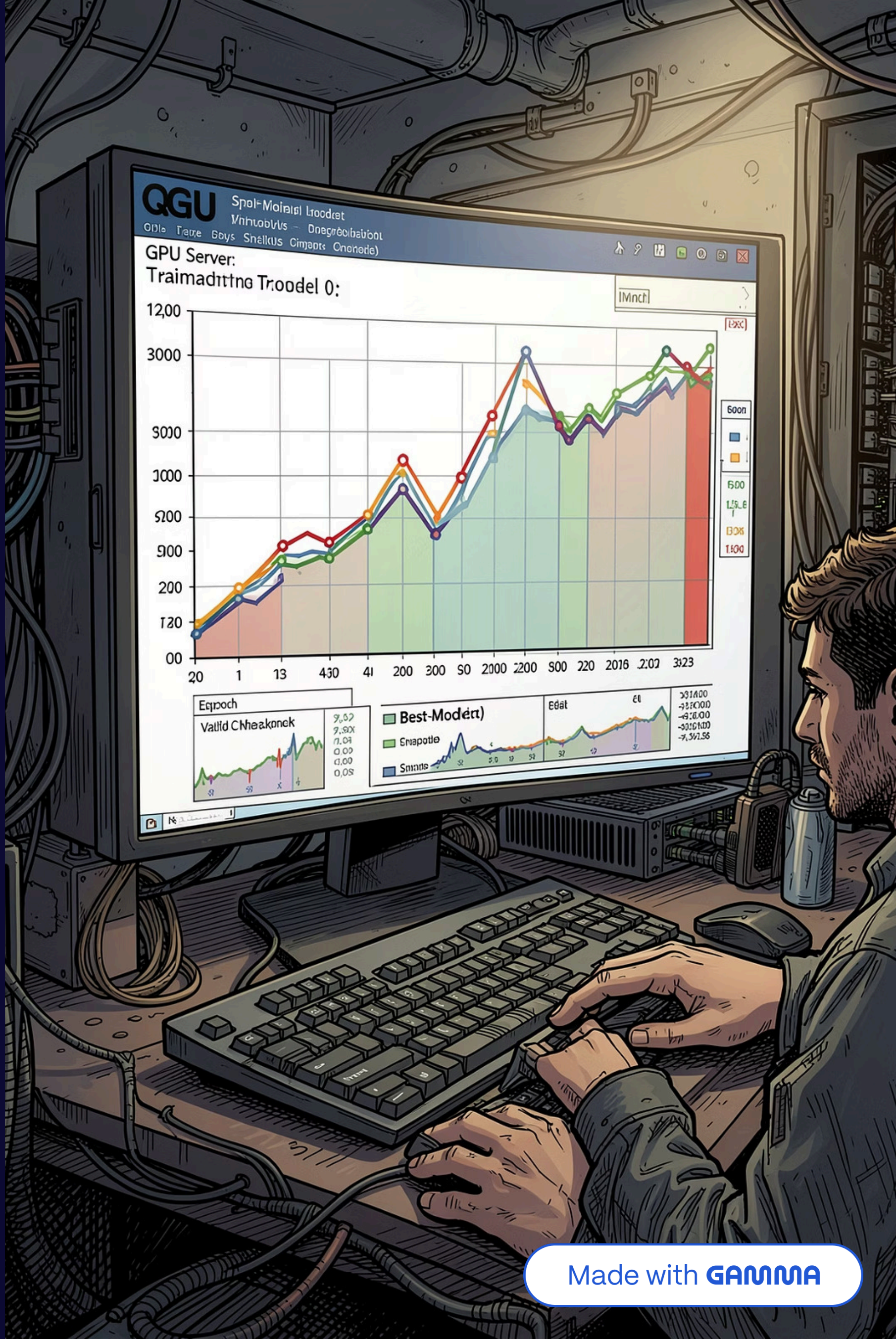
Small scans → pad with zeros

The purpose is to set the input dimensions identically.

Classifier, loss, and training loop

Classifier MLP: 320 → 128 → 64 → 1, final activation: Sigmoid(logit).
Loss: BCEWithLogitsLoss with pos_weight to account for class imbalance.

- Training loop: forward → compute loss → backpropagate → optimizer.step → scheduler step.
- Validation each epoch; checkpoint best AUC; early stop on no improvement.
- Metrics to report: AUC (primary), F1, accuracy, precision/recall. Use calibration checks for clinical use.



```
CT Scan
→ Preprocessing
→ 3D CNN
→ Image features
                \
                 → Fusion → Classifier → Cancer Prediction
                /
Tabular Data
→ MLP Encoder
→ Clinical features
```

OverView

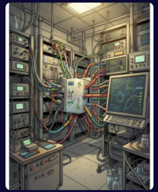
Why this design is robust — Summary & practical tips

Attention (SE) boosts interpretability

Channel-wise weighting focuses the network on diagnostically relevant features — helps when visualizing activation maps for review.

Resampling ensures physical-scale consistency

Standard voxel spacing prevents the model from learning scale artifacts and improves generalization across sites.



Multimodal fusion improves clinical relevance

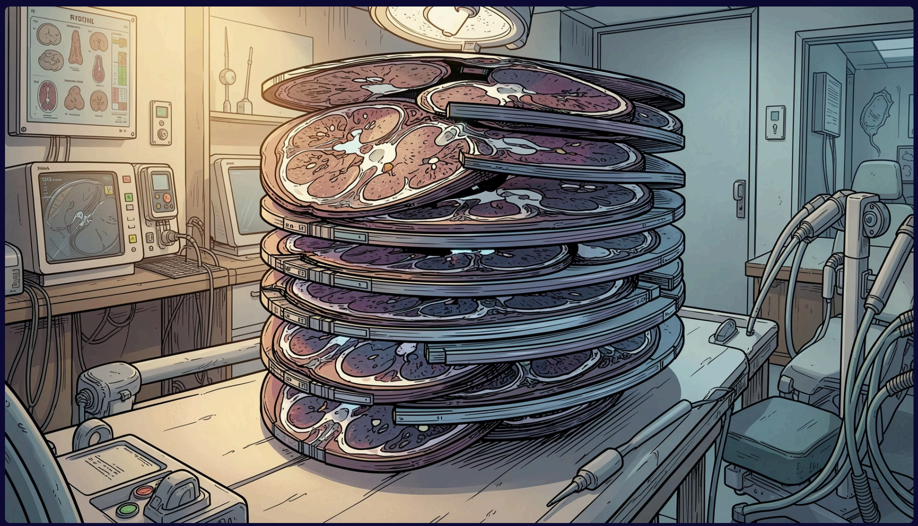
Combining imaging with patient data yields higher AUC and more clinically actionable predictions than either modality alone.

Final takeaway: Standardize inputs → preserve anatomical fidelity → use residual + attention 3D encoder → fuse with structured data → train with balanced loss and validate with AUC. This pipeline is practical, reproducible, and aligned with clinical evaluation needs.

Medical Image Approach

Part 2 — Step 4: Choose which nodule and based on what?

- LIDC-IDRI dataset characteristics: 152 patients, multi-reader annotations (up to 4 radiologists per nodule)
- Nodule characteristics recorded per annotation: malignancy, subtlety, sphericity, margin, lobulation, spiculation, texture, calcification



Selection Criteria

Patient CT Scan

↓

Parse XML annotations

↓

Keep only: unblindedReadNodule

↓

Filter valid nodules:

✓ malignancy score

✓ sufficient size

✓ valid mask ✓ ≥2 slices

↓

Choose balanced representative nodule

↓

Export: CT.nii.gz + Mask.nii.gz

Part 2 — Step 5: Masks exporting

- Reads LIDC XML annotation file, extracts per-nodule data: nodule ID, malignancy score, subtlety, sphericity, spiculation, and ROI contour points per slice
- Saves mask as `.nii.gz` with same spacing, origin, and direction as the original CT using `SimpleITK.CopyInformation`

mask__nodule_5__reader_3__mal_2__class_lo...nii.gz

Part 2- Step 6: PreProcessing

Exploratory Metadata Notebook

- Extracts metadata from NIfTI headers per case: CT shape (Z, H, W), voxel spacing in mm, HU min/max, mask presence
- **CT shape (Z, H, W)** — how many slices the scan has (depth), and how tall and wide each slice is in pixels. For example: 130 slices × 512 × 512 pixels
- **Voxel spacing in mm** — the real-world physical size of each 3D pixel. This tells you how coarse or fine the scan resolution is
- **HU min/max** — the darkest and brightest intensity values in the scan measured in Hounsfield Units. Air is around −1000 HU, soft tissue around 0–80 HU, bone around +400 HU. This confirms the scan contains valid CT intensity ranges

Part 2 - Step 6: PreProcessing

Full Preprocessing Pipeline

- **Lung segmentation:** thresholds raw HU volume at < −320 HU to isolate lung parenchyma, removes small noise objects (`min_size=500`), fills holes slice-by-slice, retains the two largest connected components (left + right lung)
- Exports lung segmentation mask as `lung_mask.nii.gz` + axial PNG per patient
- **Normalization:** clips CT volume to lung window [−1000, +400 HU] and rescales to [0, 1] float32
- Exports normalized CT as `preprocessed_volume.nii.gz` + axial, coronal, sagittal PNGs per patient
- **Tumor mask alignment:** binarizes `mask__*.nii.gz` (threshold > 0.5), resamples to CT shape using nearest-neighbor interpolation (`order=0`) to preserve binary values, applies morphological cleanup (hole filling, largest component retention)
- Exports aligned tumor mask as `tumor_mask.nii.gz` with same NIfTI affine as CT (required for PyRadiomics spatial alignment)
- Exports mask PNGs (axial, coronal, sagittal) and red-overlay PNGs for visual inspection

Step 7 — Feature Extraction

1

First-order features:

Intensity distribution statistics

(Mean, Median, Entropy, Skewness, Kurtosis, Energy)

2

Shape Features

3D tumor geometry

(Volume, Surface area, Sphericity, Elongation, Compactness)

3

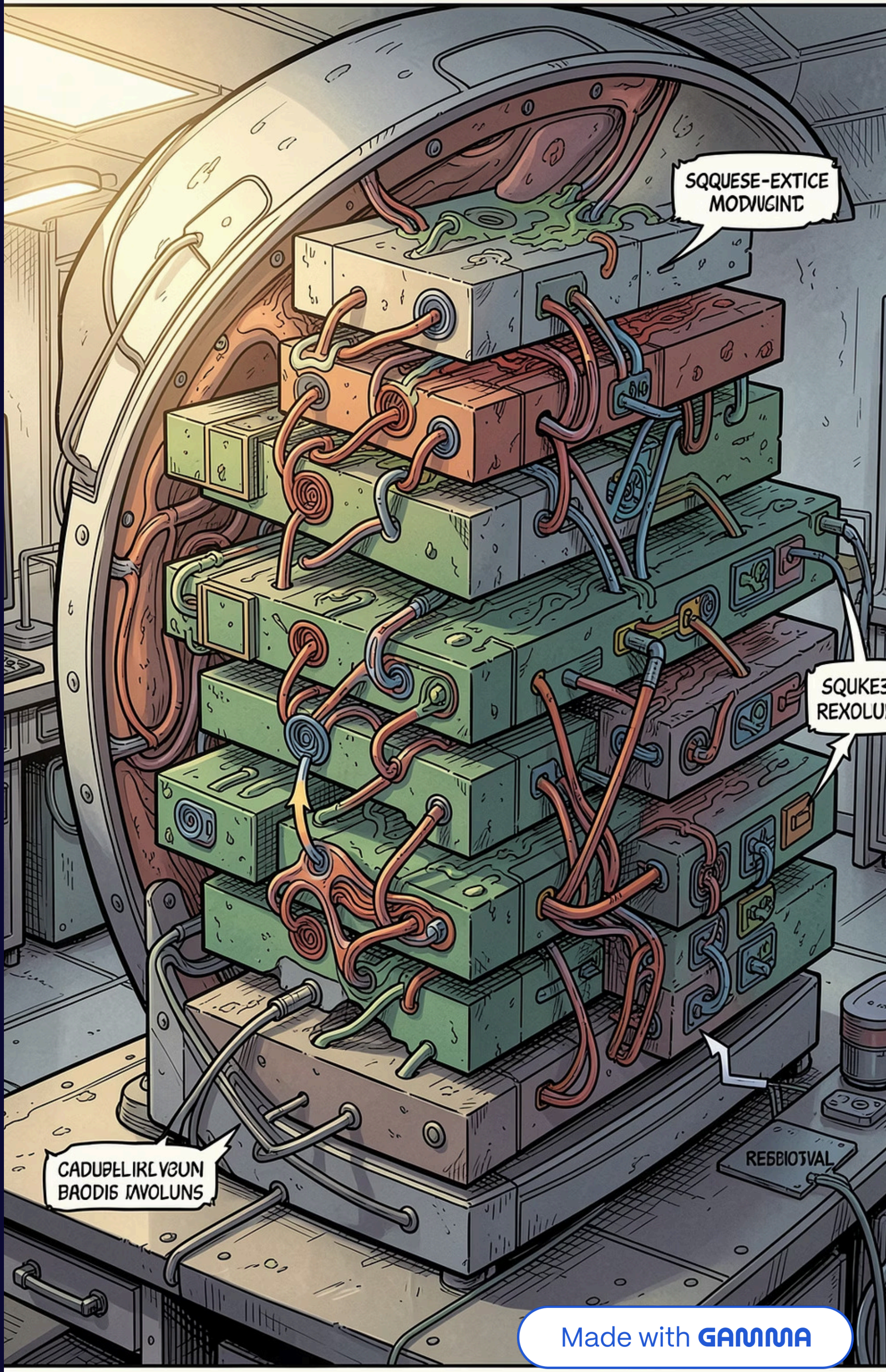
Texture Features

GLCM, GLRLM, GLSZM, GLDM, NGTDM capturing spatial heterogeneity and structural irregularity

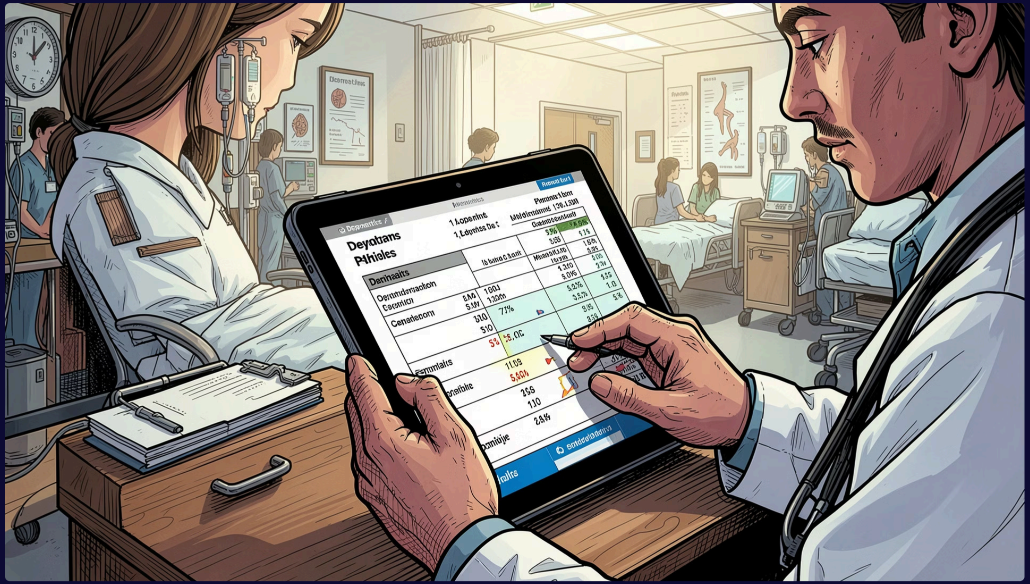
4

Filter-based features

Wavelet decomposition (multi-scale frequency), LoG (σ = 1.0, 2.0, 3.0 mm) for edge.



Step 8 - Feature Selection



Selection:

- LASSO regression (LassoCV, 5-fold cross-validation) applied to identify the most discriminative features via L1 regularisation; features with non-zero coefficients retained
- PCA applied to LASSO-selected features retaining components explaining 95% of cumulative variance
- Outputs: `features_lasso_selected.csv` (interpretable named features) and `features_pca_reduced.csv` (reduced principal components)

Last Step - Models Classification

Model Type

- Multimodal deep learning classifier combining 3D CT imaging with tabular radiomic/clinical data
- Dual-input Keras model trained to predict malignancy score as a multi-class classification problem (scores 1–5)

4-5 high risk

1-2 low risk

3 is intermediate

Tabular Preprocessing

Tabular Preprocessing (`preprocess_tabular_data`)

- Target labels (malignancy scores 1–5) encoded to integers (0–4) using `LabelEncoder`
- Missing tabular values imputed with column mean using `SimpleImputer`
- All numeric tabular features standardized to zero mean and unit variance using `StandardScaler`

Network Architecture (`build_multimodal_network`)

- **Branch 1 — Image (3D CNN):** two blocks of `Conv3D → MaxPooling3D → BatchNormalization`, then `GlobalAveragePooling3D` → `Dense(64)` — extracts spatial/volumetric tumor features
- **Branch 2 — Tabular:** `Dense(64) → Dropout(0.2) → Dense(32)` — processes radiomic and clinical features
- **Fusion:** both branches concatenated → `Dense(64) → Dropout(0.3)` → `Softmax output` with N classes
- Compiled with Adam (lr=1e-4) and sparse categorical cross-entropy loss

Training (split_and_train)

- Patient-level train/validation split (80/20)

Output

- Extracts Radiomic Features within the volume & mask
- Predicts whether patient is cancerous/non-cancerous
- Predicts malignancy class (0–4 mapped back to scores 1–5) per patient

Evaluation

- AUC-ROC per class (one-vs-rest for multi-class)
- Sensitivity and Specificity — clinically more meaningful than accuracy for cancer detection
- Confusion matrix across 5 malignancy levels
- Kaplan-Meier survival analysis linking predicted malignancy score to patient outcome

Limitations

- Small dataset (152 patients) limits generalizability
- No external validation set
- Masks being inside of xml files

Thank You!